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Survey Analysis Framework for Systematic Analysis Using AI and GEN AI Tools Paper

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Abstract

When organizations collect survey data from their stakeholders and employees to gauge their sentiments, opinions, and experiences, the analysis of mixed-format survey data, which combines structured multiple-choice questions (MCQs) and unstructured open-ended responses, poses a significant challenge in extracting scalable, actionable insights that can inform business decisions. However, traditional survey analysis methods are plagued by limitations, including the requirement for extensive manual effort, domain expertise, and disjointed workflows, which hinder the efficient extraction of meaningful insights from mixed-format data. The primary challenge lies in deriving actionable insights from mixed-format survey data, which demands a unified, AI-driven approach that can automate sentiment analysis, identify outliers, and uncover hidden patterns in survey responses. To address this challenge, this study proposes a unified, AI-driven framework that integrates classical machine learning (ML) with Generative AI (GenAI) to provide a comprehensive understanding of overall survey results including sentiments, outliers and other analytics. The synergy between machine learning and GenAI enables the framework to accurately identify patterns and trends, achieving a predictive accuracy and yielding insightful results comparable to human annotation. An evaluation of the framework's effectiveness, based on a stakeholder satisfaction survey, reveals high satisfaction rates with respect to free-text summarization quality (89%) and theme extraction efficiency (90%), demonstrating the potential of this hybrid approach to revolutionize survey research methodologies and inform evidence-based decision-making across domains such as healthcare, education, and enterprise analytics.

Keywords: Free-Text, MCQ, Large language model, Random Forest, Artificial neural network, Machine learning, Classification, GenAI

Introduction

In today's data-driven business landscape, organizations are increasingly relying on survey data to gauge the sentiments, opinions, and experiences of their stakeholders and employees, with the ultimate goal of informing strategic decisions that drive growth, improvement, and competitiveness. As a result, surveys have become an essential tool for organizations to collect valuable feedback, identify areas for improvement, and measure the effectiveness of their initiatives. However, when organizations collect survey data from their stakeholders and employees, they often encounter a significant challenge: the analysis of mixed-format survey data, which combines structured multiple-choice questions (MCQs) and unstructured open-ended responses. This mixed-format data, which is typically characterized by a combination of quantitative and qualitative elements, poses a significant challenge in extracting scalable, actionable insights that can inform business decisions, due to the complexities of integrating and analyzing both structured and unstructured data sources. Despite the importance of survey data in driving business outcomes, the lack of effective methods for analyzing mixed-format survey data can lead to incomplete, inaccurate, or misleading insights, ultimately hindering an organization's ability to make informed decisions. Therefore, there is a pressing need for innovative solutions that can effectively address the challenges associated with analyzing mixed-format survey data, and provide organizations with the actionable insights they need to drive business success.

Open-ended questions in questionnaires present vital views of the respondents' opinions, emotions, and feedback. However, interpreting extensive questionnaires not only requires tremendous time and effort, but even the complexity of open-ended questions complicates it for researchers to delve into [1]. Hence, the traditionally used methods in survey analysis require training researchers and accounting analysis by different researchers for more precise results [1][2]. This paper discusses the incorporation of Large Language Models (LLMs) into the survey analysis pipeline and how they differ from traditional machine learning approaches.

Problem Definition & Objective

Traditional survey analysis methods have several limitations, particularly when extracting meaningful insights from all data types (quantitative and qualitative), especially free-text responses. Existing tools often require manual effort, lack advanced AI capabilities, and are not optimized for mixed-format data analysis. These limitations create a gap in efficiently understanding survey results.

The problem addressed in this paper is the challenge of deriving actionable insights from mixed-format survey data, which includes both structured multiple-choice questions and unstructured open-ended responses, due to the limitations of traditional methodologies that require extensive manual effort, domain expertise, and disjointed workflows, resulting in inefficiencies in capturing nuanced user sentiments and behavioral patterns.

Research Objectives

The main objectives of this study are to:

- Develop a unified, AI-driven framework for systematic survey analysis.
- Automate the analysis of sentiments from mixed-format survey data.
- Identify outliers and uncover hidden patterns in survey responses.
- Provide actionable insights that can inform decision-making.
- Reduce manual effort, domain expertise, and disjointed workflows required for survey analysis.
- Overcome the limitations of traditional methodologies for survey data analysis.

Literature Review

This literature review was carried out through an extensive search of academic databases, conference proceedings, and industry reports, concentrating on the convergence of AI, NLP, and survey research, with a specific focus on sentiment analysis, text mining, and ML methodologies.

Generative AI for Open-Ended Survey Responses

Several paper works have utilized LLMs to complement the analysis of free-text survey questions, especially in use cases such as education, healthcare, and user research. Since traditional Machine Learning (ML) techniques are labor-intensive to interpret manually. LLMs performed well with minimal preprocessing, typically limited to prompt and query generation only. The literature indicates a growing interest in comparing LLMs with conventional ML approaches, particularly for thematic clustering and sentiment analysis.

Zhang, W. et al [3] examined the performance of models across 13 tasks on 26 datasets, with the final results compared to those of small language models (SLMs). To evaluate the models, the authors employed both few-shot and zero-shot settings. For the LLMs, the authors used the models without any additional training. In contrast, the SLMs were trained on specific datasets, which showed outstanding performance. The results demonstrate that LLMs perform well on simpler tasks in the zero-shot setting. However, there are still some limitations as LLMs strive when dealing with complex task.

Maehlum, P. et al [4] evaluated whether LLMs could accurately annotate the sentiment of open-ended patient comments, by comparing this performance with that of expert human annotators. Using a dataset of patient's responses on a medical survey from Norwegian Institute of Public Health. The authors used two open-source LLMs for Norwegian (ChatNorT5, based on T5, and NorMistral, based on Mistral) under zero-shot and prompt-induced few-shot settings. The experiments were conducted for either a 4-class or 2-class sentiment analysis task. As a result, ChatNorT5 and NorMistral demonstrated abilities close to those of humans in binary sentiment classification achieving an F1-score of 89.3% and 84.9% respectively. However, human's experts and LLMs faced a challenge in predicting neutral sentiment, often misclassifying it.

Ding, Xiuhao, et al. [5] assessed the performance of ML and LLMs in qualitative survey response analysis in studies of engineering education. The author used free-text responses collected from students' feedback on a Universal Design for Learning (UDL) project implemented via the ClassTranscribe video platform. The study contrasted both traditional ML approaches—such as Support Vector Machines (SVM), Random Forest (RF), and Latent Dirichlet Allocation (LDA)—and LLMs such as BERT, BART, and LLaMA. For sentiment analysis, BERT performed better with an accuracy of 87%, followed by SVM and RF, which achieved accuracies of 79%.

In topic modeling, the authors applied LDA to segments of questions related to social learning. In contrast, the authors used LLaMMA to generate topics on all open-ended survey questions. As a result, LDA generated lists of topics that mapped well to manually labeled themes while requiring human effort to interpret. LLaMMA missed information related to neutral or negative responses. However, in summarization, LLaMA outperformed BART by a significant margin, producing summaries that were generally as good as those produced by humans. Overall, the study finds that ML and LLM-based methods are potent tools for triangulating qualitative, hand-coded analysis. BERT is well-suited explicitly to sentiment labeling, and LLaMA shows potential for thematic summarization. However, LLMs require minimal preprocessing, which is typically limited to generating prompts and queries.

Li, A[6] 2025 compares sentiment analysis on COVID-19 survey data using LLMs and classical classification models. The surveys were collected by Stanford and National Institutes of Health (NIH) Stanford. The models considered in this comparison include: Twitter-RoBERTa-base, DeBERTa-v3-base-absa-v1.1, GPT-4o, Llama-3.3-70B-Instruct, Gemini-2.0-Flash. As a result, large language models

outperformed the traditional neural network models in sentiment analysis. When it comes to comparing the Human Emotion GPT-4o, it surpassed the other LLMS. However, LLMs require extensive computational resources to function.

NLP & Machine Learning in Survey Analysis

Natural Language Processing (NLP) and ML have become foundational in modern survey analysis, particularly for handling mixed-format survey data which includes both free-text and multiple-choice responses. Several recent studies have proved the effectiveness of various NLP and ML techniques in analyzing diverse data types.

Krishna et al. [7] conducted a study on sentiment analysis techniques applied to restaurant customer reviews using various machine learning algorithms and NLP techniques. The authors applied classifiers SVM, Naïve Bayes, Decision Trees, Random Forest, and K-Nearest Neighbors, in combination with different feature extraction techniques include Bag of Words, TF-IDF, and N-gram models. The dataset went through several preprocessing steps including noise removal, stop-word, and stemming. Their results demonstrated that the SVM classifier with N-gram features achieved the highest accuracy (95.23%), emphasizing the importance of both preprocessing and feature selection on model performance

Monika et al [8]. presented a comprehensive survey of machine learning and deep learning approaches used in sentiment analysis. The paper outlines various methodologies for sentiment analysis, including preprocessing techniques, (e.g., tokenization, stemming, lemmatization), feature extraction methods (e.g., TF-IDF, Word2Vec), and both supervised and unsupervised learning models. It also, highlights the performance of SVM, Naïve Bayes, Decision Trees algorithms, and deep learning methods like LSTM, Bi-LSTM, and Transformer networks. The Authors highlight the strengths and limitations of each approach and emphasize that deep learning models, especially Transformer-based architectures, offer promising results for sentiment classification, though they often require substantial computational resources. This survey paper provides a solid foundation for selecting appropriate sentiment analysis models based on data type, domain, and resource availability.

Lavanya et al [9]. presented a comprehensive review of deep learning methods applied to text classification tasks in social healthcare data using NLP. Their study highlights how models such as Feed forward neural networks (FFNNs) Recurrent Neural Networks (RNNs), Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRUs) , along with preprocessing and embedding techniques like Word2Vec, GloVe, and classification algorithms are used to process large-scale unstructured data like reviews, posts, and survey responses. The authors emphasize that deep learning outperforms traditional machine learning in capturing semantic relationships and context, making it effective for complex textual data like free-text survey responses.

Despite the importance of survey data in business decision-making, the existing literature reveals several shortcomings and gaps in the current methodologies and tools used for survey analysis. The literature survey highlights the need for a comprehensive framework that can effectively handle mixed-format survey data, integrating both quantitative and qualitative data analysis techniques.

Methodology

This section presents the design and architecture of the proposed Generic Analytical Framework, a modular and scalable framework for systematic survey analysis. The framework integrates traditional data science methods with advanced artificial intelligence and machine learning techniques to provide a comprehensive and automated approach to analyzing mixed-format survey data, comprising both structured and unstructured responses, and is designed to support the extraction of actionable insights and informed decision-making. The overall solution architecture is divided into four core layers: Data Upload, Preprocessing Layer, Generic Analytical Framework, and Generative AI Component.

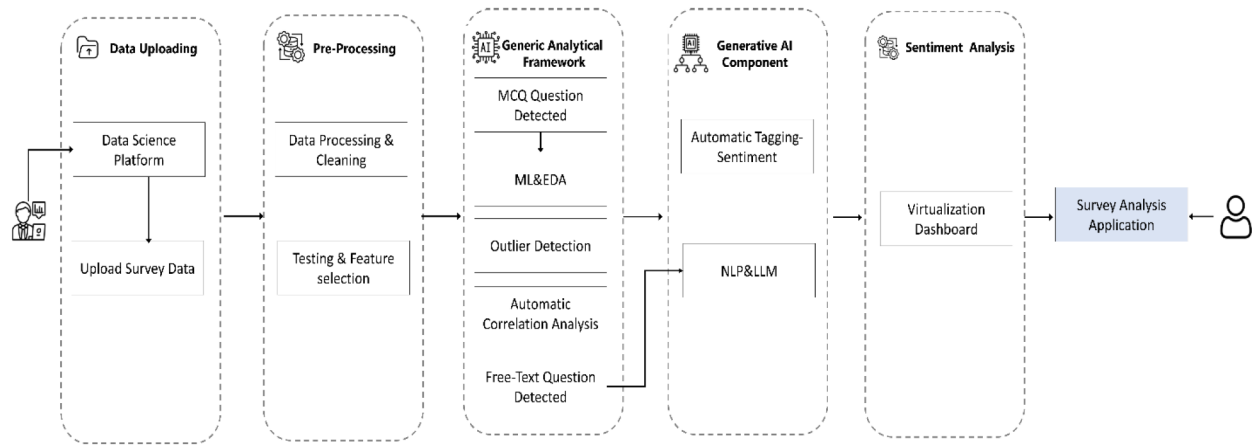


Figure 1—Architecture of the Generic Analytical Framework

Dataset and Data Preprocessing

In this framework, survey data is ingested and preprocessed using a systematic approach. The dataset is uploaded to the Data Science framework, a robust data science platform, where data normalization, deduplication, and format harmonization ensure consistency. Automated data cleaning minimizes the risk of manual bias, preparing both structured and unstructured responses for downstream machine learning and generative AI-driven insights.

Data Ingestion. The survey data is uploaded to the Data Science framework, initiating the data preprocessing pipeline. This step involves two primary processes:

1. **Data Science Platform:** Provides a robust environment for data storage, processing, and analysis.
2. **Upload Survey Data:** The survey data is uploaded to Data Science framework, making all relevant data available for subsequent processing and analysis.

Data Preprocessing. Data preprocessing steps are crucial in preparing the survey data for analysis. These steps involve:

1. **Data Processing and Cleaning:** Missing values are handled, duplicates are removed, and data normalization is performed to ensure data consistency and reliability.
2. **Testing Feature and Selection:** The most relevant features that contribute to model performance are selected, optimizing the data for analysis.

Generic Analytical Framework

The Generic Analytical Framework is a crucial component of the proposed approach, enabling the analysis of survey data to extract valuable insights. This section describes the framework's components, including exploratory data analysis (EDA) and demographic profiling, as well as feature mapping and correlation modeling.

Exploratory Data Analysis and Demographic Profiling. To evaluate the quality, structure, and distribution of survey responses, the framework automatically performs an exploratory data analysis (EDA) and demographic profiling. This automated process utilizes advanced AI tools to stratify the data sample based on demographic variables, revealing subgroup-level differences in response patterns and identifying variations in behavioral trends, satisfaction scores, and engagement levels across population segments. The framework's built-in outlier detection mechanism, which employs Z-score analysis and threshold-based filtering, automatically identifies departments or groups with extreme or unrealistic values in significant variables, such as satisfaction scores and engagement metrics, that fall below a predefined threshold.

(e.g., 70% positive sentiment). For instance, if the users of a particular department exhibit sentiment scores below the threshold, the framework automatically flags this department as an outlier, enabling targeted interventions. Furthermore, the automated EDA process generates relevant insights, including interactive graphs, charts, and tables, which provide a comprehensive understanding of the survey data. These visualizations and summaries are automatically tailored to the specific needs of the analysis, ensuring that stakeholders can quickly and easily grasp the key findings and trends in the data. By automating the EDA and demographic profiling process, the framework enables rapid and accurate identification of areas for improvement, facilitating data-driven decision-making and strategic action.

Automated Feature Mapping and Correlation Modeling. The framework is equipped with a mechanism to automatically run through a classification model, eliminating the need for manual exploratory data analysis (EDA). This automated process employs supervised machine learning techniques, including ensemble-based models such as Random Forest (RF) and Artificial Neural Networks (ANNs), to identify the most influential predictors among the given set of target variables. These techniques can comprehend convoluted non-linear relationships between features and targets and derive meaningful importance scores.

The classification model is designed to automatically reveal the important features and contributing questions for the survey, providing stakeholders with a clear understanding of the key drivers of the survey outcomes. The framework's automated mechanism ensures that the classification model is run iteratively, refining the results and providing a comprehensive understanding of the survey data. This automated process eliminates the need for manual EDA, reducing the time and effort required to analyze the survey data and providing stakeholders with rapid and accurate insights.

The output of the classification model includes a set of importance scores, which indicate the contribution of each feature to the predicted outcome. These importance scores can be used to identify the most critical questions in the survey, enabling stakeholders to focus on the areas that have the greatest impact on the target variables. The framework's automated mechanism also provides a detailed report of the classification model's results, including the predicted outcomes, importance scores, and contributing questions, providing stakeholders with a comprehensive understanding of the survey data and enabling them to make informed decisions.

To extract valuable insights from the survey data, the framework employs a robust feature engineering process, which includes feature mapping and correlation analysis. Feature mapping is performed to transform discrete, qualitative factors in the MCQ section into numerical feature representations through ordinal encoding, categorical index mapping, and domain-specific grouping, thereby preserving the semantic structure. This enables the framework to capture the nuances of the survey data and prepare it for advanced analytics.

Next, correlation coefficients, particularly Pearson's correlation, are used to quantify the direction and magnitude of linear relationships between input features. However, to further identify the most influential predictors among the given set of target variables, a classification model is employed. The classification model plays a crucial role in revealing the important features that contribute to the target variables, enabling the framework to identify the key drivers of the survey outcomes.

Generative AI Component

Open-ended questions in surveys realize evasive, qualitative thoughts that are essential to obtain for a general sense of user opinion and experience. To process this unstructured text data, the second component of our solution leverages transformer-based LLMs. This layer conducts prompt-based inference for sentiment, thematic extraction, and summarization. Through integration of these findings with the structured dataset, the system enables multi-dimensional analysis that not only quantifies behavior but also interprets the emotional and contextual drives behind user feedback. The Generative AI Component utilizes advanced LLMs to analyze unstructured open-ended responses and derive actionable insights. This component enables

a comprehensive understanding of both behavioral patterns and user sentiment, providing a more detailed analysis of the survey data.

Sentiment and Theme Analysis. Generative AI inference is operated on free-text responses from the survey using state-of-the-art LLMs to identify central concerns, user intentions, and recurring topics, as well as classify polarity labels (positive, neutral, or negative) by contextual semantic classification based on the LLM's power to perform nuanced interpretation of subjective text. This approach is utilized under zero-shot and few-shot settings. The application of LLMs enables the extraction of valuable insights from unstructured data, providing a more comprehensive understanding of user sentiment and themes.

Summarization and Semantic Representation. Concise summaries are generated using LLMs to entail the complete emotions and key stories from the entire set of text responses. Additionally, actionable recommendations are identified, either given or implicitly suggested, within the user feedback. The recommendations include ways to improve usability, enhance content clarity, and fulfill unsatisfied user demands. The combination of summarization and suggestion extraction provides decision-makers with strategic input, enabling them to be aware of user sentiment on a considerable scale and plot feedback into actionable items for possible enhancement. This approach enables unsupervised exploration of user sentiment and thematic distribution beyond what is highlighted in MCQ responses.

The proposed Generic Analytical Framework combines traditional statistical examination and advanced language models to unlock full insights from questionnaires. Supervised learning, correlation statistics, and feature engineering were applied to analyze structured questions and identify meaningful patterns of behavior and predictors. Meanwhile, open-ended responses were processed by LLMs for sentiment classification, theme identification, summarization, and suggestion extraction. It permitted the balanced analysis of quantitative trends and qualitative remarks, supporting a multidimensional understanding of user responses.

Result and Evaluation

This section presents the results of the evaluation of our proposed AI-driven framework for systematic survey analysis. We assess the performance of our framework in terms of its ability to accurately analyze sentiments, identify outliers, and uncover hidden patterns in survey responses. We also evaluate the framework's efficiency, scalability, and usability, using a combination of quantitative and qualitative metrics. The results are based on a comprehensive evaluation of our framework using a large dataset of survey responses, and are compared to traditional methodologies to demonstrate the effectiveness and advantages of our approach.

Classification Model Evaluation

This section explains the performance of the classification model for identifying the most influential features used for informing data-driven decisions and optimizing the survey instrument. To evaluate the classification model, the accuracy metrics was used.

Table 1—Model Accuracy for Predicting Target Outcomes

Model	Accuracy
Random Forest	67
Artificial Neural Network (ANN)	67.5

Both the RF and ANN models achieved an accuracy of 67% on the validation set, indicating comparable performance in capturing the underlying patterns within the structured survey data. Even if the model's accuracy is 67%, the feature importance scores can still provide valuable insights into the relationships

between the input features and the target variables. The scores can help identify the most critical features that drive the predicted outcomes, which can be used to inform decision-making and optimize the survey instrument.

Manual Coding vs LLM-Based Analysis

Traditional manual coding of open-ended survey responses is a labor-intensive process that involves human analysts reading each response, assigning sentiment labels (positive, neutral, or negative), and categorizing them into predefined themes. While this method provides accurate insights, it is not only time-consuming but also resource-intensive, often requiring weeks to analyze large datasets. This prolonged timeline can delay decision-making and reduce the agility of organizations in responding to feedback.

Our study employed a Large Language Model (LLM) to analyze survey responses, offering a powerful alternative to manual coding. Using zero-shot and few-shot prompting, the LLM automatically classified sentiments and themes with minimal training data, reducing processing time from weeks to seconds per response. Our experimental results showed that the LLM achieved high accuracy in capturing sentiments, demonstrating its robustness and efficiency. Additionally, the model successfully handled responses in multiple languages, including Arabic and Saudi dialects, further showcasing its versatility and capability in diverse linguistic contexts. While potential biases were considered, rigorous testing and human validation ensured the accuracy and reliability of the LLM's output.

In conclusion, the LLM-based approach offers enhanced efficiency, accuracy, and scalability compared to manual coding, making it a valuable tool for survey analysis. Its ability to process multilingual responses adds an extra layer of functionality, highlighting its potential for future applications in global survey research.

Stakeholder Satisfaction Survey

To validate the framework's practical utility, a stakeholder satisfaction survey was conducted among roughly 100 domain experts , data scientists, and decision-makers who interacted with the framework's outputs. The survey assessed six dimensions of usability and impact, with responses aggregated on a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree).

Table 2—Satisfaction Survey

Survey Dimensions	Sample Questions
Ease of Use	How intuitive is the framework's interface for uploading and analyzing survey data
Insight Clarity	How effectively does the framework translate complex data into actionable insights?
Time Efficiency	To what extent does the framework reduce manual effort compared to traditional analysis methods
Customization	How well does the framework adapt to domain-specific survey designs (e.g., healthcare, education)?
Actionability	How useful are the sentiment summaries and thematic recommendations for decision-making?"
Overall Satisfaction	Rate your overall experience with the framework's performance and reliability.

Table 3—Evaluation of the framework's

Aspect	Result
Quantitative - Free-Text Responses	89% satisfaction
Quantitative - MCQ	95% of stakeholders highlighted efficiency
GenAI Sentiment Analysis	Users praised the zero-shot classification for capturing nuanced feedback without any retraining.

Conclusion and Future Direction

In this paper, a novel AI-driven framework presented for systematic survey analysis, designed to efficiently process mixed-format survey data and provide actionable insights. Our framework leveraged the strengths of both traditional machines learning and generative AI techniques to automate the analysis of sentiments, identify outliers, and uncover hidden patterns in survey responses. The results of our evaluation demonstrated the effectiveness of our framework in extracting valuable insights from survey data, with high accuracy and efficiency.

The proposed framework has significant implications for various fields, including healthcare, education, and enterprise feedback analytics, where survey data is commonly used to inform decision-making. By automating the analysis of survey data, our framework can help organizations reduce manual effort, improve response rates, and gain a deeper understanding of their customers, patients, or students.

While the framework has shown promising results, there are several avenues for future research and improvement. One potential direction is to explore the application of the framework to other types of text data, such as social media posts or customer reviews. Additionally, plans include investigating the use of transfer learning and multi-task learning to further improve the accuracy and efficiency of the framework.

Another important area of future research is the development of more sophisticated visualization tools and techniques to facilitate the interpretation and communication of survey results. This could involve the use of interactive dashboards, heat maps, or other data visualization methods to help stakeholders quickly understand the key findings and insights from the survey data.

Finally, the framework is envisioned to be integrated with other AI-powered tools and technologies, such as natural language generation and chatbots, to create a more comprehensive and automated survey analysis platform. This could enable organizations to not only analyze survey data more efficiently but also to respond to customer feedback and concerns in a more timely and personalized manner.

In conclusion, the AI-driven framework for systematic survey analysis has the potential to revolutionize the way organizations collect, analyze, and act on survey data. This work is expected to have a significant impact on the field of survey research and look forward to continuing to refine and improve our framework in the years to come.

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